

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks available |          |          |          |          |          |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 11       | (a) | (i)  |  | $2\text{Fe} + 3\text{Br}_2 \rightarrow 2\text{FeBr}_3$<br>award (1) for correct formulae of reactants<br>award (1) for correct formula of product<br>award (1) for correct balancing only if all formulae are correct                                                                                                                                                                                                                                |                 | 3        |          | 3        |          |          |
|          |     | (ii) |  | add silver nitrate (solution) (1)<br>iron(III) bromide gives a cream precipitate (1)<br>iron(II) iodide gives a yellow precipitate (1)<br><br>accept other sensible test e.g.<br>add chlorine (water) (1)<br>iron(III) bromide gives orange colour (1)<br>iron(II) iodide gives brown/dark colour (1)<br><br>add sodium hydroxide (solution) (1)<br>iron(III) bromide gives a brown precipitate (1)<br>iron(II) iodide gives a green precipitate (1) | 2               | 1        |          | 3        |          | 3        |
|          | (b) |      |  | $\frac{19.4}{19} : \frac{16.3}{80}$ (1)<br><br>$0.204 : 1.021$ (1)<br><br>$1 : 5 \text{ ratio} \Rightarrow \text{BrF}_5$ (1)<br>award (2) for correct answer with no working<br>ecf possible<br>award (2) for $\text{Br}_5\text{F}$ (where $A_r$ divided by mass with working shown)<br>award (1) for $\text{Br}_5\text{F}$ with no working                                                                                                          |                 | 3        |          | 3        | 3        |          |
|          |     |      |  | <b>Question 11 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>2</b>        | <b>7</b> | <b>0</b> | <b>9</b> | <b>3</b> | <b>3</b> |